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Exponentially weighted moving mean–variance with portfolio-level aggregation: Genetic algorithm tuning and out-of-sample evidence

Kumsong Jo , Sunhak Kim and Bongnam Ri

*Department of Financial Management,
Faculty of Finance, Kim Il Sung University Taesong District,
Pyongyang, Democratic People's Republic of Korea*

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Abstract

This paper proposes an Exponentially Weighted Moving Mean–Variance (EMMV) model that discounts older data with a forgetting factor. Unlike traditional asset-level smoothing, the EMMV aggregates portfolio-level moments across rolling windows, yielding stable, adaptive allocations. Hyperparameters are selected objectively via a genetic algorithm with walk-forward cross-validation. Using 30 U.S. stocks from 2010 to 2025, the model achieves a buy-and-hold Sharpe ratio of 1.49 — more than double that of the equally weighted portfolio — and a rebalanced Sharpe ratio of 0.69 after transaction costs, outperforming all benchmarks. Robustness analysis confirms consistently high performance over a wide parameter range, and bootstrap tests verify significant or near-significant improvements. The framework provides a practical, data-driven tool for dynamic portfolio selection in nonstationary markets.

Keywords: Mean–variance (MV) model; the exponentially weighted moving mean–variance (EMMV); Sharpe ratio; genetic algorithm (GA).

1. Introduction

Portfolio selection is a highly appealing area for financial researchers and practitioners. Since Markowitz introduced modern portfolio theory in 1952 and further refined it in 1959, the concepts of stochastic mean and variance have become widely accepted as fundamental measures of return and risk for portfolios.

*Corresponding author.

Email addresses: *18801097345@163.com

K. Jo, S. Kim & B. Ri

This framework helps investors strike a desirable balance between risk and return. Numerous researchers, including Leung *et al.* (2001); Liu *et al.* (2003); Shao *et al.* (2025), and Şen *et al.* (2025), have developed various models that treat stochastic variance as a measure of risk. They worked to ensure that modern portfolio theory aligns more closely with real-world scenarios. Leung *et al.* (2001) tried to improve the efficiency of portfolios under the future movement of stock prices by combining forecasts and Liu *et al.* (2003) proposed a Mean–Variance (MV) skewness model for portfolio selection with transaction cost. Shao *et al.* (2025) applied the MV and Mean-Expected Tail Loss (ETL) portfolio optimization techniques on earnings forecasting and robust regression-based composite models. The problem of mean–variance portfolio selection regularized with an l_0 -penalty term to control the sparsity of the portfolio was considered by Şen *et al.* (2025). Furthermore, the practical application of the mean–variance framework in diverse market conditions continues to be explored. For instance, González-Bueno *et al.* (2025) conducted a comparative performance analysis of mean–variance optimization in the emerging Colombian capital market, highlighting both its utility and the challenges posed by nonstationary environments.

Adding constraints to the standard mean–variance model makes the portfolio selection problem mathematically more complex to solve. As a result, many researchers focus on artificial intelligence to find optimal portfolios. Neural networks are among the most powerful machine learning tools (Khan *et al.*, 2021b, 2022) and have made significant advancements in recent years, particularly in the areas of financial data analysis and signal processing (Adamu, 2019; Gao and Su, 2020; Khan *et al.*, 2021a; Troiano *et al.*, 2018). The use of computational methods, such as machine learning and data analytics, to enhance traditional finance practices is becoming a widely adopted trend among leading investment firms (Allen *et al.*, 2021; Goldstein *et al.*, 2019; Rebentrost *et al.*, 2018; Sharma *et al.*, 2024; Tran *et al.*, 2019; Tsantekidis *et al.*, 2021). Yaman and Dalkılıç (2021) proposed a hybrid approach to the cardinality constraint portfolio selection problem based on a nonlinear neural network and genetic algorithm (GA). A novel neural network-based mean–variance–skewness model for optimal portfolio selection was proposed by Yu *et al.* (2008), integrating various forecasts, trading strategies, and investors' risk preferences. Cui and Zhang (2025) proposed a multi-period mean–variance portfolio selection method with real constraints based on machine learning. Adosoglou *et al.* (2021) introduced a novel investment strategy called the Semantic Similarity Portfolio (SSP), which leverages neural network embeddings.

Modern portfolio theory faces significant challenges, leading to skepticism among portfolio managers. This skepticism arises from the theory's assumption that all portfolio returns are symmetric and that future movements will follow the distribution patterns observed in historical data.

EMMV with portfolio-level aggregation

Numerous researchers have utilized fuzzy set theory to capture asset returns and investor subjectivity. Fuzzy set theory (Zadeh, 1965) addresses linguistic expressions and vague information, making it suitable for financial modeling. The portfolio decision-making problem has been extensively studied in fuzzy environments. Recent advances continue to highlight its value:

- Wang *et al.* (2023) developed a multi-criteria fuzzy portfolio selection model based on three-way decisions and cumulative prospect theory, integrating advanced decision-making frameworks with behavioral finance.
- Similarly, Yadav *et al.* (2023) proposed a multi-objective sustainable financial portfolio selection model under an intuitionistic fuzzy framework, addressing modern priorities like sustainability.
- Other scholars have also contributed significantly to this domain (García *et al.*, 2020; Gong *et al.*, 2021; Gupta *et al.*, 2021; etc.).

Researchers have estimated portfolio returns and risks using probability and credibility measures within these fuzzy sets, leading to the optimization of fuzzy portfolio selection models.

Despite these advancements, a persistent difficulty in applying the mean–variance framework to real markets is that it treats all historical data as equally informative. In reality, financial markets are nonstationary: the relevance of information decays over time, and observations from distant regimes often reflect conditions that no longer prevail. While Exponentially Weighted Moving Averages (EWMAs) are the standard tool for discounting older data, the conventional way of introducing them into portfolio choice — smoothing individual asset returns before solving a single Markowitz problem — can amplify estimation errors and produce concentrated, fragile allocations.

This paper, therefore, asks two practical questions. First, does aggregating portfolio-level moments across multiple rolling windows, and exponentially weighting recent windows more heavily, yield more stable and better-performing portfolios than the traditional asset-level smoothing? Second, can the key hyperparameters — the window size, the step between windows, and the forgetting factor — be chosen in a principled, data-driven way rather than being set arbitrarily?

To answer these questions, we construct an Exponentially Weighted Moving Mean–Variance (EMMV) model that computes a complete efficient frontier for each overlapping window of the historical data and then temporally aggregates these portfolio-level moments with exponential weights. A genetic algorithm with walk-forward cross-validation selects the window size, the lag, and the forgetting factor directly from the estimation period, eliminating the need for *ad-hoc*

K. Jo, S. Kim & B. Ri

parameter choices. The performance of the resulting strategy is evaluated against the static Markowitz model, a simple moving-window average, a traditional EWMA-MV benchmark that smooths asset returns, and the naïve equally weighted portfolio, using 30 U.S. stocks over the period 2010–2025. Both buy-and-hold and quarterly rebalanced implementations are examined, the latter with proportional transaction costs. Robustness is assessed through a sensitivity heat-map that spans a wide range of hyperparameter values, and statistical inference is performed with a moving-block bootstrap (MBB) test that accounts for time-series dependence.

The remainder of this paper is organized as follows. Section 2 presents the dynamic portfolio framework, distinguishing the proposed portfolio-level temporal aggregation from conventional asset-level smoothing and defining the benchmark models. Section 3 describes the empirical study, including the genetic-algorithm-based hyperparameter selection, the in-sample efficient frontiers, the out-of-sample buy-and-hold and rebalancing results, the robustness analysis, and the statistical hypothesis tests. Section 4 concludes with the main findings and their practical implications.

2. Portfolio Selection Models

2.1. Standard mean–variance model

The standard mean–variance model, introduced by Markowitz (1952, 1959), is a constrained optimization problem. Its two primary formulations are

Minimum Variance for a Target Return: The investor seeks the portfolio with the lowest possible risk that achieves a specified minimum expected return, τ .

$$\text{minimize } \sum_i \sum_j x_i x_j \sigma_{i,j} \quad (1)$$

$$\text{subject to } \sum_i x_i \mu_i \geq \tau, \quad (2)$$

$$\sum_i x_i = 1, \quad (3)$$

$$x_i \geq 0. \quad (4)$$

Maximum Return for a Target Risk: The investor seeks the portfolio with the highest possible return that does not exceed a specified maximum risk (variance) budget, φ .

$$\text{maximize } \sum_i x_i \mu_i \quad (5)$$

EMMV with portfolio-level aggregation

$$\text{subject to } \sum_i \sum_j x_i x_j \sigma_{i,j} \leq \varphi, \quad (6)$$

$$\sum_i x_i = 1, \quad (7)$$

$$x_i \geq 0, \quad (8)$$

where x_i is investment proportion of i th portfolio asset, μ_i is statistical expected return of i th portfolio asset, $\sigma_{i,j}$ is covariance of i th, j th portfolio assets.

For the empirical analysis in this paper, we will use the minimum variance for a target return formulation (Eqs. (1)–(4)). We will solve this problem for a range of target returns (τ) to trace out the entire efficient frontier.

2.2. Motivating dynamic modeling

The standard Markowitz model (Eqs. (1)–(8)) is inherently static: it treats the entire historical record as equally informative and assumes that the distribution of returns is stationary. In reality, financial markets are nonstationary — the relevance of information decays over time, and observations from distant regimes may no longer reflect current conditions. A practically useful portfolio model should therefore place greater weight on recent data and discount obsolete information.

The EWMA, popularized by RiskMetrics (J.P. Morgan/Reuters, 1996), is the standard tool for such discounting. However, the way EWMA is applied to portfolio selection admits two fundamentally different routes.

The traditional approach smooths the individual asset return series itself, recursively updating a time-varying covariance matrix Σ_t and then solving a single Markowitz optimization with the final estimate Σ_T . While widely used, this asset-level smoothing can amplify estimation errors when the covariance matrix is ill-conditioned, often producing concentrated portfolios with artificially low perceived risk.

Our proposal is portfolio-level temporal aggregation. Instead of smoothing the raw returns, we partition the entire history into overlapping rolling windows of size m shifted by l days. Within each window, we compute the portfolio-level moments — the mean portfolio returns and the portfolio variance — for a given weight vector. These window-specific moments are then aggregated using exponential weights governed by a forgetting factor $\lambda \in (0, 1]$. Intuitively, each window provides a noisy snapshot of the efficient frontier; the exponential weighting forms a temporal ensemble that filters out the noise while adapting to structural changes. The resulting EMMV estimator yields more stable, better diversified portfolios.

Thus, the proposed EMMV model generalizes the Markowitz framework. When $\lambda = 1$, the estimator becomes a simple arithmetic average of the window-specific

K. Jo, S. Kim & B. Ri

portfolio moments. As the number of windows N grows large — i.e., when the step size l is small and the window size m covers a substantial part of the history — this average converges to the unconditional full-sample moments, recovering the standard static model in the limit. This convergence is analogous to the way the mean of many sample-average estimates approaches the population means. In the degenerate case $m = T$ and $l = 1$, only a single window exists, and the EMMV model immediately collapses to the standard Markowitz solution.

2.3. Simple moving mean and variance models

Building upon the theoretical foundation of dynamic estimation outlined in the previous section, we now propose a moving-window framework that replaces the single unconditional estimate of portfolio moments with a collection of window-specific conditional estimates. These models are designed to be more responsive to recent market trends by incorporating a forgetting mechanism, which will be fully developed in Sec. 2.4.

2.3.1. Construction of rolling windows

Let the full return series consist of T daily observations for n assets. When the stock price return time series of the k th subset is denoted by r_t^k , the expected return and variance of the k th subset of an individual stock can be expressed as

$$\mu_{r^{(k)}} = E[r_t^k], \quad (9)$$

$$\sigma_{r^{(k)}}^2 = E[r_t^k - \mu_{r^{(k)}}]^2. \quad (10)$$

For any given subset k , we can compute the conditional expected return and variance, analogous to the full-sample estimates in the standard model, but based only on the data within the window $[k, k + m - 1]$: The number of l lagged m -size subset ($l \leq m$) among the total time series of stock price return (r_T) is $N = \lfloor \frac{T-m}{l} \rfloor + 1$, where $\lfloor \cdot \rfloor$ least integer function. The observations of k th subset and $(k+1)$ th subset is $r^{(k)} = [r_k r_{k+1}, \dots, r_{k+m-1}]$ and $r^{(k+1)} = [r_{k+l}, r_{k+l+1}, \dots, r_{k+l+m-1}]$, respectively, with $l, m \in \mathbb{N}$ constant. The first index ($k = 1$) corresponds to the oldest window and the last index ($k = N$) to the most recent one.

2.3.2. Portfolio-level moments within each window

Within each window k , we estimate the sample mean vector $\boldsymbol{\mu}^{(k)} = (\mu_{r_1^{(k)}}, \dots, \mu_{r_n^{(k)}})^T$ and the sample covariance matrix $\boldsymbol{\Sigma}^{(k)}$ with elements $\rho_{i,j}^{(k)} \sigma_i^{(k)} \sigma_j^{(k)}$. For a given portfolio weight vector $\mathbf{X} = (x_1, \dots, x_n)^T$, the portfolio return

EMMV with portfolio-level aggregation

and portfolio variance attributable to the k th window are then

$$r_p^{(k,l,m)} = \sum_i x_i \mu_{r_i}^{(k,l,m)}, \quad (11)$$

$$\sigma_{p^{(k,l,m)}}^2 = \sum_i \sum_j x_i x_j \rho_{i,j}^{(k,l,m)} \sigma_i^{(k,l,m)} \sigma_j^{(k,l,m)}. \quad (12)$$

Each pair $(r_p^{(k)}, \sigma_{p^{(k)}}^2)$ can be interpreted as a noisy snapshot of the efficient frontier that would be obtained if only the data in window k were available.

2.3.3. Simple arithmetic aggregation

The Simple Moving Mean–Variance (SMMV) model aggregates the window-specific portfolio moments by taking their arithmetic average:

$$\mu_{r_p^{(s)}} = \frac{1}{N} \sum_{k=1}^N r_p^{(k,l,m)}, \quad (13)$$

$$\sigma_{p^{(s)}}^2 = \frac{1}{N} \sum_{k=1}^N \sigma_{p^{(k,l,m)}}^2. \quad (14)$$

Here, $\mu_{r_p^{(s)}}$ is the simple moving mean of portfolio returns, $\sigma_{p^{(s)}}^2$ is the simple moving variance of portfolio returns, and N is the number of subsets. Because the portfolio variance $\sigma_{p^{(k,l,m)}}^2$ is a quadratic form in the weight vector $\mathbf{X}$, the SMMV objective can equivalently be written as

$$\sigma_p^{2(s)}(\mathbf{X}) = \mathbf{X}^T \bar{\Sigma} \mathbf{X}, \quad \text{where} \quad \bar{\Sigma} = \frac{1}{N} \sum_{k=1}^N \Sigma^{(k)}. \quad (15)$$

Thus, the SMMV optimal portfolio for a target return τ is obtained by solving the standard quadratic program with the averaged moments $\boldsymbol{\mu}_{\text{SMMV}} = \frac{1}{N} \sum_k \boldsymbol{\mu}^{(k)}$ and $\bar{\Sigma}$.

2.4. Exponentially weighted moving mean–variance proposed model

2.4.1. Exponential weighting of portfolio moments

To move beyond a simple average and assign more weight to recent conditions, we introduce an exponential forgetting factor $\lambda \in (0, 1]$. This recursive formulation is chosen for its computational efficiency and its foundation in stochastic gradient descent and adaptive filter theory, providing a smooth decay of influence consistent with many models of financial time series memory.

K. Jo, S. Kim & B. Ri

The exponential weight on the mean value of k th subset is defined by the recursive formula:

$$\omega_{k,\lambda} = \lambda\omega_{k-1,\lambda} + 1. \quad (16)$$

This equation recursively calculates the sum of the weights used in the exponential weighting scheme up to the k th subset, ensuring the weights are properly normalized in the mean calculation (Eq. (17)), where $\omega_{k,\lambda}$ is the exponential weight on the mean of k th subset, λ the forgetting factor, $\omega_{k-1,\lambda}$ is the exponential weight on the portfolio mean of $(k+1)$ th subset.

Figure 1 illustrates the impact of the forgetting factor on the moving mean of portfolio returns, which determines the weight assigned to past data. A forgetting factor of 0.9 places more emphasis on older data compared to a forgetting factor of 0.1. Meanwhile, a forgetting factor of 1.0 indicates infinite memory, meaning that all previous samples receive equal weight.

In the exponential weighting method, the moving mean is computed recursively using the following formula:

$$E_{r_p^{(k)}} = \left(1 - \frac{1}{\omega_{k,\lambda}}\right)E_{r_p^{(k-1)}} + \frac{1}{\omega_{k,\lambda}}r_p^{(k)}, \quad (17)$$

where $E_{r_p^{(k)}}$ is the cumulative exponential moving mean up to subset k , $r_p^{(k)}$ is the portfolio return computed on the k th subset (Eq. (11)), and the multipliers $\frac{1}{\omega_{k,\lambda}}$ and

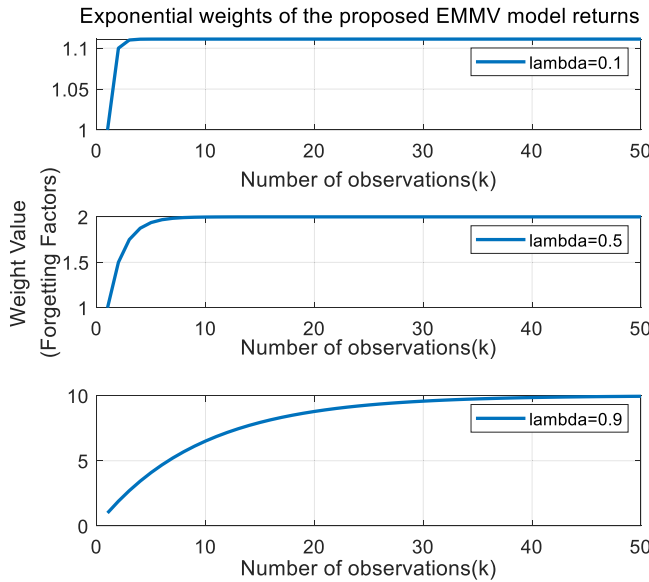


Fig. 1. The impact of forgetting factor $\lambda = 0.1, 0.5, 0.9$ on the moving mean weight. The weights are calculated using Eq. (16) for $k = 1$ to 50. k is the number of observations.

EMMV with portfolio-level aggregation

$1 - \frac{1}{\omega_{k,\lambda}}$ ensure that the most recent subset receives weight proportional to $\frac{1}{\omega_{k,\lambda}}$ while the past mean is smoothly down-weighted.

For the portfolio variance, the exponential weight is defined by

$$v_{k,\lambda} = \frac{2\lambda(1 - \lambda^{k-1})}{(1 - \lambda)(1 + \lambda)}, \quad (18)$$

where $v_{k,\lambda}$ is the normalizing factor that guarantees the exponentially weighted variance estimator remains unbiased under the assumption of a locally constant true variance. Its behavior is depicted in Fig. 2.

Figure 2 shows the impact of the forgetting factor on the moving variance of portfolio return, which determines how much weight past data is given. As the age of the data increases, the magnitude of the weighting factor decreases exponentially and approaches zero without ever reaching it. In other words, the recent variance has more influence on the current variance than the older variance.

The exponentially weighted moving variance is then

$$S_{k,\lambda}^2 = \frac{1}{v_{k,\lambda}} \sum_{i=1}^k \lambda^{k-i} \sigma_{p^{(i)}}^2, \quad (19)$$

where $\sigma_{p^{(i)}}^2$ is the portfolio variance on the i th subset (Eq. (12)). Note that recent subsets (large i) receive larger multipliers λ^{k-i} , thereby making the estimator more responsive to current market conditions.

The EMMV framework nests both the SMMV and the classical Markowitz model as limiting cases.

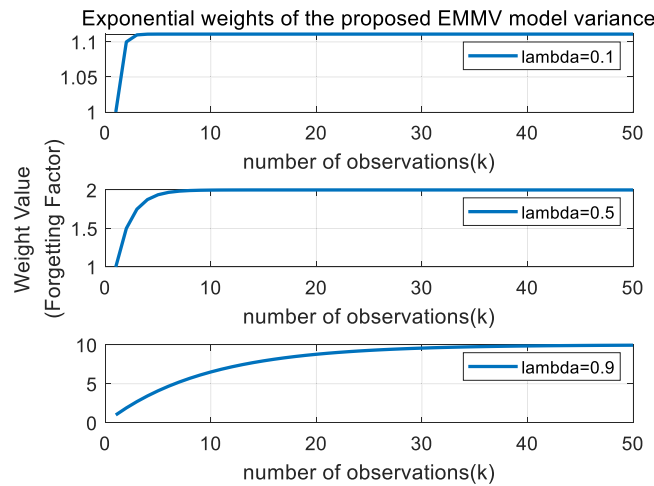


Fig. 2. The impact of forgetting factor $\lambda = 0.1, 0.5, 0.9$ on the moving variance weight. The weights are calculated using Eq. (18) for $k = 1$ to 50. k is the number of observations.

K. Jo, S. Kim & B. Ri

- When $\lambda = 1$, Eqs. (16)–(19) reduce to equal weights across all subsets, and the EMMV collapses to the SMMV described in Sec. 2.3.
- As the number of subsets $N = \lceil \frac{T-m}{l} \rceil + 1$ grows large — i.e., when the step size l is small and the window size m covers a substantial part of the history — the law of large numbers implies that the equally-weighted average of the subset-specific moments converges to the unconditional full-sample moments. Hence, for $\lambda = 1$ and sufficiently large N , the EMMV (and SMMV) recover the standard Markowitz solution.
- In the extreme case $m = T$ and $l = 1$, only a single subset exists, and the EMMV immediately collapses to the static model regardless of λ .

Thus, the EMMV is a proper generalization of the Markowitz framework, introducing a tunable degree of adaptivity through λ while retaining the classical model as a special case.

2.4.2. EMMV-optimal portfolios

To obtain investable portfolios, the EMMV moments are plugged into a standard mean–variance optimization. When the complete efficient frontier is of interest, the minimum-variance-for-a-given return-problem becomes, for a target return τ ,

$$\text{minimize} \left(S_{N,\lambda}^2 = \frac{1}{v_{N,\lambda}} \sum_i^N \lambda^{N-i} \sigma_{p^{(i)}}^2 \right) \quad (20)$$

subject to

$$E_{r_p^{(N)}} = \left(1 - \frac{1}{\omega_{N,\lambda}} \right) E_{r_p^{(N-1)}} + \frac{1}{\omega_{N,\lambda}} r_p^{(N)} \geq \tau, \quad (21)$$

$$\sum x_i = 1, \quad (22)$$

$$x_i \geq 0 \quad (23)$$

and the dual maximum-return-for-a-given-risk formulation is obtained by interchanging the objective and constraint as in Eqs. (24)–(27). In the empirical sections of the paper, we focus on the maximum Sharpe ratio portfolio, which is obtained by maximizing the ratio of the EMMV expected return to the square root of the EMMV variance under the same full-investment and long-only constraints.

$$\text{maximize} \left(E_{r_p^{(N)}} = \left(1 - \frac{1}{\omega_{N,\lambda}} \right) E_{r_p^{(N-1)}} + \frac{1}{\omega_{N,\lambda}} r_p^{(N)} \right) \quad (24)$$

subject to

$$S_{N,\lambda}^2 = \frac{1}{v_{N,\lambda}} \sum_i^N \lambda^{N-i} \sigma_{p^{(i)}}^2 \leq \varphi, \quad (25)$$

EMMV with portfolio-level aggregation

$$\sum x_i = 1, \quad (26)$$

$$x_i \geq 0. \quad (27)$$

2.5. Traditional EWMA-MV (benchmark)

To isolate the benefit of portfolio-level temporal aggregation, we implement a conventional asset-level EWMA-MV model that follows the RiskMetrics methodology (J.P. Morgan/Reuters, 1996). The expected return vector $\boldsymbol{\mu}_t$ and covariance matrix $\boldsymbol{\Sigma}_t$ are updated daily from the return vector $\mathbf{r}_t$:

$$\boldsymbol{\mu}_t = (1 - \lambda_{\text{ewma}})\mathbf{r}_t + \lambda_{\text{ewma}}\boldsymbol{\mu}_{t-1}, \quad (28)$$

$$\boldsymbol{\Sigma}_t = (1 - \lambda_{\text{ewma}})\mathbf{r}_t\mathbf{r}_t^T + \lambda_{\text{ewma}}\boldsymbol{\Sigma}_{t-1}, \quad (29)$$

with the industry-standard decay factor $\lambda = 0.94$. The final estimates $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$ are used as inputs to the standard Markowitz program (Eqs. (1)–(4)) to obtain the optimal portfolio.

The traditional model smooths individual asset returns before optimization; the proposed EMMV smooths portfolio moments across multiple windows. This design difference has important practical consequences. Asset-level smoothing tends to produce ill-conditioned covariance matrices that favor highly concentrated portfolios with artificially low perceived risk (see Sec. 3.2, where the traditional EWMA-MV frontier is confined to a deceptively narrow low-risk region). In contrast, the EMMV temporally aggregates entire efficient frontiers, yielding more diversified and stable allocations. By including both approaches in the empirical comparison, we can cleanly measure the contribution of portfolio-level temporal aggregation.

3. Empirical Results and Comparison

3.1. Data description and experimental setup

Our empirical analysis utilizes daily price data for 30 stocks listed on major U.S. exchanges, spanning from January 1, 2010, to September 1, 2025. The selected tickers are: NVDA, MSFT, AAPL, JPM, MA, GS, UNH, LLY, ISRG, AMZN, HD, MCD, NFLX, TMUS, DIS, GE, CAT, RTX, COST, PG, PM, XOM, CVX, VLO, NEE, SRE, DUK, SPG, PLD, AMT. This universe spans ten GICS sectors: Information Technology (NVDA, MSFT, AAPL, MA), Financials (JPM, GS), Healthcare (UNH, LLY, ISRG), Consumer Discretionary (AMZN, HD, MCD), Communication Services (NFLX, TMUS, DIS), Industrials (GE, CAT, RTX), Consumer Staples (COST, PG, PM), Energy (XOM, CVX, VLO), Utilities (NEE,

K. Jo, S. Kim & B. Ri

SRE, DUK), and Real Estate (SPG, PLD, AMT). This broad cross-section provides a diverse test bed for portfolio strategies.

The 2010–2025 sample period deliberately spans several distinct macroeconomic regimes: the post-Global Financial Crisis recovery (2010–2019), the COVID-19 pandemic shock (2020), the unprecedented monetary and fiscal stimulus era (2020–2021), and the aggressive interest-rate hiking cycle (2022–2024). By design, this long and turbulent window challenges any portfolio strategy with severe nonstationarity, making it an ideal testbed for the adaptive capability of the proposed EMMV framework.

The dataset is split into

- Estimation (In-Sample) period: January 1, 2010–August 31, 2024 (approximately 3,700 trading days).
- Testing (Out of Sample) period: September 1, 2024–September 1, 2025 (1 year, approximately 252 trading days).

The Out-of-Sample period is deliberately chosen to be recent and unseen by any model during training, including the hyperparameter tuning phase.

3.1.1. Hyperparameter selection via genetic algorithm

The three hyperparameters (l, m, λ) of the EMMV model must be chosen objectively to avoid arbitrary tuning. To this end, we employ a GA that optimizes the parameters using only the In-Sample data, maintaining a strict separation from the test period. The search space and GA configuration are summarized in Algorithm 1, and the subsequent paragraph reports the convergence outcome.

In our experiment, the bounds were set to $l \in [1, 21]$, $m \in [21, 300]$, $\lambda \in [0.01, 0.99]$. The GA was configured with population size $P = 100$, maximum generations $G = 50$, elite count $E = 5$, crossover fractions $c = 0.8$, and mutation rate $\mu = 0.1$. The algorithm was executed with a fixed random seed for reproducibility.

The GA converged to $l^* = 1$, $m^* = 83$, $\lambda^* = 0.2344$, corresponding to an average cross-validation Sharpe ratio of $\overline{SR} = 1.6899$. This relatively low forgetting factor indicates that the model responds aggressively to recent market movements, while the 83-day window provides sufficient history for stable covariance estimation. Importantly, as demonstrated in Sec. 3.5, the out-of-sample performance is remarkably insensitive to variations around these optimal values, confirming the robustness of the chosen hyperparameters.

Figure 3 illustrates the convergence behavior of the GA over 50 generations. The solid line tracks the best fitness value (i.e., the negative of the highest average validation Sharpe ratio) found up to each generation, while the dashed line shows the mean fitness of the entire population.

*EMMV with portfolio-level aggregation***Algorithm 1.** Genetic algorithm for EMMV hyperparameter selection

Input: In-sample daily return matrix $\mathbf{R}_{\text{train}} \in \mathbb{R}^{T \times N}$,
 Lower/upper bound $[l_{\min}, m_{\min}, \lambda_{\min}], [l_{\max}, m_{\max}, \lambda_{\max}]$,
 population size P , max generations G , crossover fraction c , mutation rate μ

Output: Optimal hyperparameters (l^*, m^*, λ^*)

1. Initialization

Generate an initial population of P individuals, each representing a vector $[l, m, \lambda]$ uniformly within the bounds, with l and m constrained to integers.

2. Fitness Evaluation

For each individual $[lm\lambda]$:

- a. Round l and m to the nearest integer.
- b. Perform Walk-forward cross-validation on $\mathbf{R}_{\text{train}}$:
 - Define minimum training length $T_{\min} = 500$ days, validation window $T_{\text{val}} = 63$ days.
 - For each validation step $t = T_{\min} + 1 : T_{\text{val}} : (T - T_{\text{val}})$:
 Train an EMMV portfolio using data up to day $t - 1$ to obtain weights w .
 Compute the daily portfolio returns on the validation window $[t, t + T_{\text{val}} - 1]$
 Calculate the annualized Sharpe ratio SR_t of this validation window.
 Compute the average validation Sharpe ratio $\overline{\text{SR}} = \frac{1}{K} \sum_t \text{SR}_t$.
 Assign fitness $f = -\overline{\text{SR}}$ (minimization problem).

3. Genetic Operations

Repeat for G generations:

- a. **Selection:** Tournament selection retaining the best E elites.
- b. **Crossover:** Random crossover of surviving individuals with fraction c .
- c. **Mutation:** Gaussian perturbation with adaptive feasible mutation, rate μ .
- d. Re-evaluate fitness for the new population.

4. Termination

Stop after G generations or when the best fitness stagnates for 10 consecutive generations.

5. Output

Return the individual $[l^* m^* \lambda^*]$ with the lowest final fitness, i.e., the highest $\overline{\text{SR}}$.

The best fitness steadily improves from approximately -1.65 ($\text{SR} \approx 1.65$) in the first few generations to -1.75 ($\text{SR} \approx 1.75$) at termination, with most of the gain occurring within the first 20–25 generations. Thereafter, the curve plateaus, indicating that the algorithm has located a stable optimum and that further generations yield only marginal refinement. The mean fitness closely follows the best value, signaling that the population remains consistently concentrated around

K. Jo, S. Kim & B. Ri

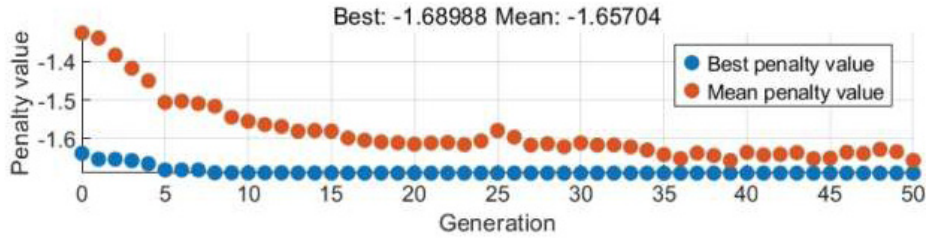


Fig. 3. Convergence of the genetic algorithm for hyperparameter selection.

high-quality solutions rather than dispersing erratically. The absence of sudden jumps or late-stage deterioration confirms that the GA did not overfit to the validation folds and that the selected hyperparameters ($l = 1, m = 83, \lambda = 0.2344$) represent a genuine, robust optimum of the cross-validation objective.

3.1.2. Benchmark models

To assess the incremental value of the proposed EMMV framework, we compare it against four benchmark strategies that span the spectrum from fully static to industry-standard dynamic approaches. All models are estimated on the same In-Sample period and are subject to the identical portfolio constraints (full investment, long-only). Table 1 summarizes their main characteristics.

SMMV (Sec. 2.3) replaces the full-sample moments with the arithmetic average of moments computed over overlapping rolling windows of size $m = 83$ and lag $l = 1$. It is essentially the EMMV model with $\lambda = 1$, i.e., no forgetting.

Traditional EWMA-MV applies exponential smoothing directly to the asset-return series (RiskMetrics, $\lambda = 0.94$). It operates at the asset level, in contrast to the proposed EMMV, which aggregates moments at the portfolio level.

Table 1. Summary of benchmark models.

Model	Estimation	Hyperparameters	Adaptivity
Standard MV	Full-sample moments	None	None
SMMV	Simple moving window average	$l = 1, m = 83$	Window-based
Traditional EWMA-MV	Exponential smoothing on asset returns	$\lambda = 0.94$	Asset-level decay
Naïve 1/N	None	None	None
EMMV (proposed)	Exponential smoothing on portfolio-window moments	$l = 1, m = 83, \lambda = 0.2344$	Portfolio-level decay

EMMV with portfolio-level aggregation

Naïve $1/N$ allocates equal weight to all $N = 30$ assets. It requires no estimation and often outperforms optimized models due to the absence of estimation error (DeMiguel *et al.*, 2009).

The EMMV (proposed) model is configured with the GA-tuned parameters ($l = 1, m = 83, \lambda = 0.2344$) and performs exponential smoothing on portfolio-window moments.

All portfolio optimizations were solved using MATLAB's Portfolio class with the TrustRegionCP solver.

3.2. In-sample performance: Efficient frontiers

Figure 4 displays the efficient frontiers of the five strategies computed from the In-Sample period (January 2010–August 2024). It is important to recognize that the expected return and portfolio risk are constructed from different estimators for each model. Consequently, the absolute coordinates of the frontiers are not directly comparable: a portfolio that appears to offer a higher expected return for a given risk on one frontier does not necessarily dominate a portfolio on another frontier when evaluated under a common out-of-sample measure.

Nevertheless, the shape and slope of each frontier convey valuable information about the risk-return trade-off that the model perceives *ex-ante*. All frontiers are

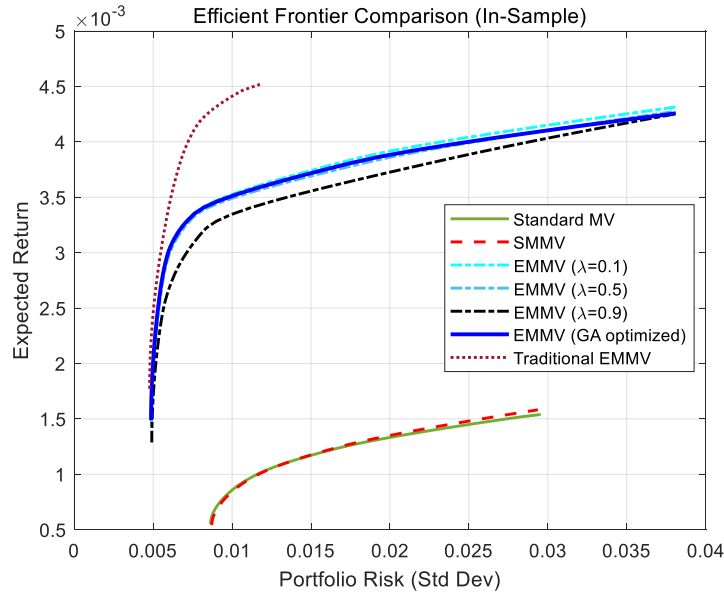


Fig. 4. Efficient frontiers — Standard MV, SMMV, traditional EWMA-MV, EMMV ($\lambda = 0.1, 0.5, 0.9$), EMMV (GA optimized $\lambda = 0.2344$).

K. Jo, S. Kim & B. Ri

smooth and differentiable, confirming that each model yields a well-posed optimization problem. The crucial observation is the steepness of the trade-off, which reflects the attainable Sharpe ratio.

The EMMV frontiers — particularly the one associated with the GA-optimized forgetting factor $\lambda = 0.2344$ — exhibit a markedly steeper ascent than those of the static and traditional benchmarks. This means that, for a small increase in perceived risk, the model projects a disproportionately large increase in expected return. In other words, the EMMV formulation generates a much higher implied maximum Sharpe ratio than the standard MV, SMMV. Under the GA-tuned hyperparameters ($m = 83, l = 1$), the SMMV efficient frontier proved to be nearly identical to that of the standard MV. This is expected: with many overlapping windows, the equally-weighted average of window-specific moments converges to the unconditional full-sample moments. The GA-optimized $\lambda = 0.2344$ yields the steepest slope among the EMMV variants, indicating that this parameter value extracts the most favorable risk-return trade-off from the historical data.

The traditional EWMA-MV frontier also has a relatively steep slope, reflecting the benefit of exponential smoothing at the asset level. However, its feasible risk range is substantially narrower than that of the EMMV frontiers. Because the asset-level EWMA covariance matrix tends to produce highly concentrated portfolios with artificially suppressed volatility, the traditional model is confined to a deceptively narrow low-risk region and fails to span a broader, more realistic risk spectrum. In contrast, the EMMV frontiers span a wider spectrum of risk, enabling the construction of portfolios that range from very conservative to more aggressive allocations while still maintaining a superior Sharpe ratio.

In summary, although the absolute positions of the frontiers are not directly comparable — because each model measures expected return and risk with its own distinct estimator — the in-sample curves convey two clear messages. First, the EMMV frontiers are markedly steeper than those of the standard and traditional benchmarks, implying a substantially higher ex-ante Sharpe ratio as perceived by the model. Second, the EMMV frontiers span a far broader and more realistic range of portfolio risk, whereas the traditional EWMA-MV frontier is squeezed into a deceptively narrow low-risk interval. The GA-tuned EMMV with $\lambda = 0.2344$ delivers the steepest trade-off among all variants. Whether this ex-ante superiority materializes as genuine out-of-sample outperformance is examined in the following sections.

3.3. Out-of-sample performance and comparison

The practical merit of a portfolio strategy is judged by its performance on data that were entirely excluded from estimation and hyperparameter tuning. We therefore

EMMV with portfolio-level aggregation

evaluate all models on the test period from September 2024 to September 2025 (approximately 252 trading days). For each model, we construct the portfolio that maximizes the in-sample Sharpe ratio using the full estimation window (January 2010–August 2024), and then track its behavior out-of-sample. Two implementation modes are examined: a static buy-and-hold setting and a quarterly rebalancing setting that accounts for proportional transaction costs.

3.3.1. Buy-and-hold comparison

In the buy-and-hold scenario, the initial optimal weights are held unchanged throughout the test year, incurring no trading costs. Table 2 reports the annualized performance of all four models, and Figure 5 traces their cumulative wealth trajectories.

The EMMV portfolio clearly dominates all benchmarks. It achieves a Sharpe ratio of 1.49, more than double that of the naïve equally weighted portfolio (0.71) and over four times the standard Markowitz portfolio (0.35). Its annualized return of 25.15% is the highest in the panel, yet its volatility (16.92%) is only slightly above that of the naïve and MV portfolios, indicating that the EMMV model genuinely improves the risk-return trade-off rather than simply amplifying risk.

Figure 5 reinforces this message with a single glance. The cumulative return curve of the EMMV portfolio (solid blue) lies consistently above the other three lines throughout the entire test year. The separation begins early and widens steadily, with the EMMV ending the period at approximately 24%, while the traditional EWMA-MV (dotted green) reaches about 23% and the naïve 1/ N (dashed red) roughly 10%. The standard MV portfolio (dash-dot magenta) is barely distinguishable from the horizontal axis, confirming that the unconditional Markowitz solution collapses out-of-sample.

The traditional EWMA-MV is the closest competitor, yet it still trails the proposed EMMV by roughly 10% in the Sharpe ratio. Both models employ exponential forgetting, but the traditional approach smooths individual asset returns, which can produce concentrated and unstable portfolios when the covariance matrix becomes ill-conditioned. The EMMV, by contrast, smooths portfolio-level

Table 2. Out-of-sample buy-and-hold performance (September 2024–September 2025).

Model	Annualized return	Annualized volatility	Sharpe ratio
EMMV (proposed)	25.15%	16.92%	1.4865
Naïve 1/ N	11.28%	15.92%	0.7086
Standard MV	4.63%	16.47%	0.3469
Traditional EWMA-MV	22.23%	13.34%	1.3501

K. Jo, S. Kim & B. Ri

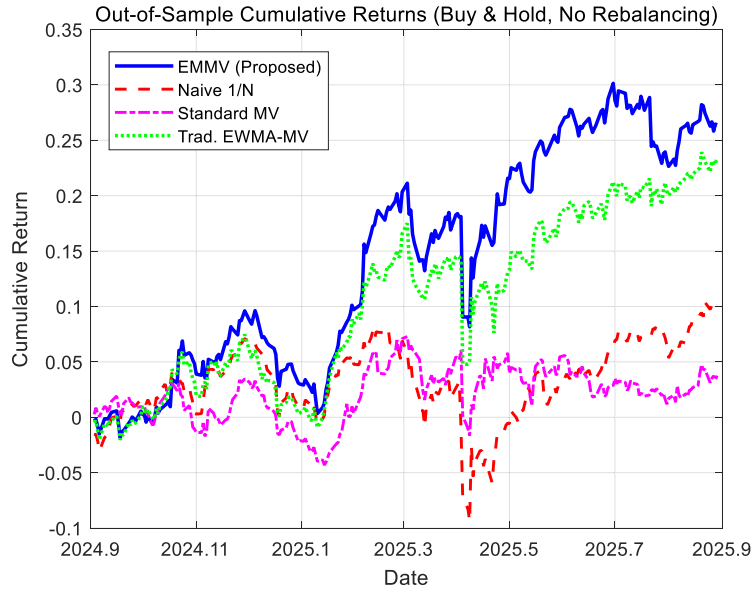


Fig. 5. Cumulative returns of the buy-and-hold portfolios over the test period (September 2024–September 2025).

moments over multiple rolling windows, yielding more diversified allocations that better survive the transition to out-of-sample data.

In sum, the buy-and-hold results establish a clear performance ranking: EMMV $\gg$ Traditional EWMA-MV $>$ Naïve $1/N$ $\gg$ Standard MV. The EMMV model not only offers the highest absolute return but also the highest reward per unit of risk, validating the GA-tuned forgetting mechanism and the portfolio-level aggregation design. The following section examines whether this advantage persists when portfolios are periodically rebalanced in the presence of transaction costs.

3.3.2. Dynamic rebalancing with transaction costs

To evaluate the practicality of the strategies under realistic trading conditions, we simulate a quarterly (63-trading-day) rebalancing process over the test period. The choice of this frequency is directly motivated by the hyperparameter selection procedure: the genetic algorithm (Sec. 3.1.1) maximizes the average Sharpe ratio over a series of 63-day validation windows. Using the same horizon for out-of-sample rebalancing ensures consistency between the objective that guided parameter optimization and the implementation of the resulting strategy.

At each rebalancing date, portfolio weights are recomputed using all data available up to that date (expanding window), and a proportional transaction cost of 10 basis points (0.1%) is applied to the absolute change in weights. The first

EMMV with portfolio-level aggregation

Table 3. Out-of-sample rebalanced performance (63-day rebalancing, 10 bp proportional transaction cost).

Model	Sharpe ratio	Annualized return	Annualized volatility	Total turnover	Total transaction cost
EMMV (proposed)	0.6891	13.83%	20.07%	2.98	0.30%
Traditional EWMA-MV	0.4348	8.28%	19.04%	3.47	0.35%
Standard MV	−0.3830	−7.30%	19.06%	1.31	0.13%
Naïve 1/N	0.7086	11.28%	15.91%	0.00	0.00%

day's return of each holding period is reduced by the cost, mimicking a realistic drag on performance. The Naïve 1/N portfolio, which never rebalances, serves as a zero-cost benchmark. Table 3 reports the annualized performance and turnover metrics, while Fig. 6 displays the cumulative wealth trajectories after costs.

The EMMV model delivers the highest annualized return (13.83%) and the best Sharpe ratio (0.69) among all strategies that rebalance. Its turnover (2.98) is 15% lower than that of the traditional EWMA-MV (3.47), reflecting the portfolio-level smoothing that produces more stable allocations and thus preserves more of the gross return after costs. The standard MV, by contrast, suffers a negative Sharpe ratio (−0.38) and a net loss of 7.30%, as its unconditional optimization amplifies estimation errors when rolled forward.

Although the cost-free Naïve 1/N portfolio achieves a marginally higher Sharpe ratio (0.71) owing to its lower volatility, its terminal wealth trails the EMMV's by 2.55 percentage points. Thus, for investors who prioritize return enhancement over pure volatility reduction, the EMMV remains the superior choice even net of trading costs. Figure 6 visualizes these trajectories: the EMMV curve (solid blue) consistently leads the other strategies, ending the year approximately 2 % ahead of the naïve portfolio and well above the traditional EWMA-MV and standard MV.

3.4. Robustness and sensitivity analysis

A key concern with any parametric model is sensitivity to hyperparameter choices. To address this, we constructed a grid of window sizes $m \in [50, 100]$ (step 5) and forgetting factors $\lambda \in [0.05, 0.3]$ (step 0.02), and for each pair we trained the EMMV model on the full In-Sample period and evaluated its buy-and-hold Out-of-Sample Sharpe ratio. Figure 7 displays the results as a color-coded heatmap.

The heatmap reveals a broad high-performance plateau: for $m \in [70, 90]$ and $\lambda \in [0.10, 0.30]$, the Sharpe ratio exceeds 1.3, and in many cells surpasses 1.4.

K. Jo, S. Kim & B. Ri

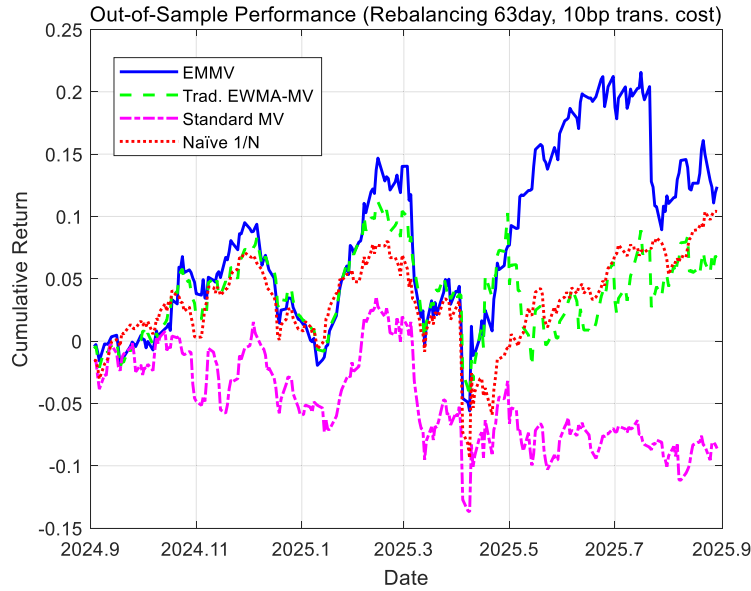


Fig. 6. Cumulative returns of the quarterly rebalanced portfolios over the test period, net of proportional transaction costs (10 bp per rebalance).

The GA-selected combination achieves the overall maximum of 1.51, but the surrounding region yields almost identical performance, demonstrating that the model's success does not hinge on a single finely-tuned parameter set. This directly addresses the criticism of arbitrary parameter selection and proves the practical robustness of the EMMV framework.

3.5. Statistical hypothesis tests

To evaluate whether the observed out-of-sample Sharpe ratio differences are statistically meaningful, we conduct an MBB test following Ledoit and M Wolf (2008). The MBB procedure resamples blocks of consecutive daily returns, thereby preserving the serial dependence structure typically present in financial time series. Compared with an i.i.d. bootstrap or asymptotic normal approximations, this approach provides more reliable inference under autocorrelation and conditional heteroskedasticity.

We set the block length to 5 trading days and generate 10,000 bootstrap replications. For each replication, we compute the annualized Sharpe ratio difference between the proposed EMMV portfolio and each benchmark portfolio. Since the research hypothesis is directional — namely that EMMV delivers superior risk-adjusted performance — we report one-sided bootstrap p -values.

EMMV with portfolio-level aggregation

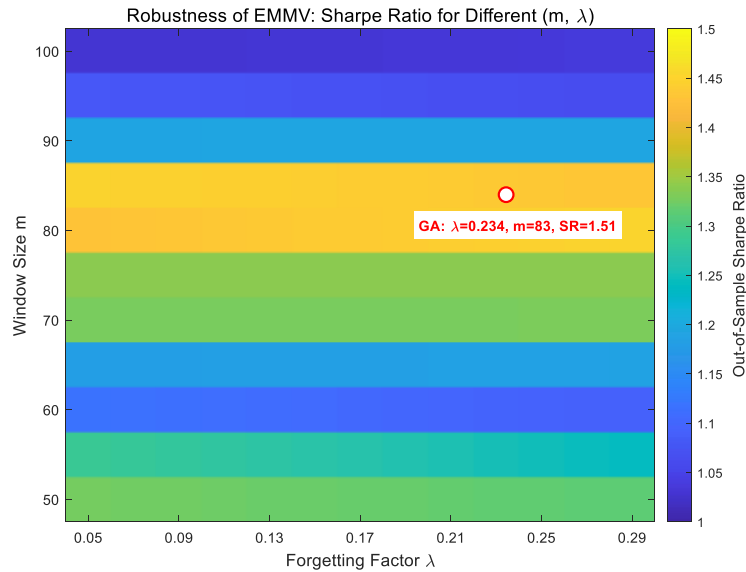


Fig. 7. Sensitivity of EMMV performance to window size and forgetting factor.

Table 4 summarizes the Sharpe ratio differentials, the corresponding one-sided p -values, and the bootstrap confidence intervals for the differences.

The bootstrap results provide supportive evidence that the proposed EMMV strategy improves risk-adjusted performance relative to the benchmark models. In particular, the EMMV portfolio exhibits a Sharpe ratio improvement of 1.1396 over the standard Markowitz portfolio, with a one-sided p -value of 0.0422, indicating statistical significance at the 5% level under the Ledoit–Wolf bootstrap framework. The improvement relative to the naïve $1/N$ benchmark is economically large ($\Delta Sharpe = 0.7779$) and is marginally significant ($p = 0.1005$), suggesting that the outperformance is meaningful although statistical power is limited by the short test horizon. In contrast, the Sharpe ratio advantage over the traditional EWMA-MV benchmark is modest ($\Delta Sharpe = 0.1364$) and not statistically significant ($p = 0.3167$).

Table 4. Moving-block bootstrap test for equality of annualized Sharpe ratios (buy-and-hold, one-sided).

Comparison	Diff (SR)	p -Value (one-sided)	95% CI
EMMV–Naïve $1/N$	0.7779	0.1005	[−1.9332, 1.2623]
EMMV — Standard MV	1.1396	0.0422	[−1.5453, 1.3141]
EMMV — Trad. EWMA-MV	0.1364	0.3167	[−0.9947, 0.7981]

K. Jo, S. Kim & B. Ri

Overall, the hypothesis testing results suggest that the EMMV framework delivers consistent and economically substantial gains in out-of-sample Sharpe ratio, particularly relative to the static Markowitz benchmark. While the one-year out-of-sample period constrains the ability to obtain strong statistical significance across all pairwise comparisons, the magnitude of the Sharpe improvements, together with the robustness plateau documented in Sec. 3.4, supports the conclusion that the EMMV approach provides practically meaningful enhancements in dynamic portfolio selection.

3.6. Discussion and limitations

The empirical results clearly favor the EMMV framework. By forgetting at the level of entire rolling windows rather than individual asset returns, the proposed model avoids the concentrated portfolios and artificially narrow risk spectra that characterize the traditional EWMA-MV approach. The genetic algorithm selected a short effective memory ($m = 83$ days, $\lambda = 0.2344$), well-suited to the turbulent Out-of-Sample year: the buy-and-hold Sharpe ratio reached 1.49, more than double that of the equally weighted portfolio, and remained at 0.69 after quarterly rebalancing with transaction costs. The robustness heatmap confirmed that this success is not tied to a single parameter combination — a broad plateau of Sharpe ratios exceeding 1.3 extends across a wide range of window sizes and forgetting factors. The MBB test provided formal support, with a significant improvement over the static Markowitz model ($p = 0.042$) and a near-significant edge over the naïve portfolio ($p = 0.101$).

Several limitations should be noted. First, the empirical evidence is drawn from U.S. equities only; broader asset classes and international markets remain to be tested. Second, the 63-day rebalancing frequency, while consistent with the GA validation setup, deserves deeper investigation under varying cost structures. Third, the tendency of the traditional EWMA-MV model to produce a narrow, low-risk spectrum is empirically documented here, but a full analytical explanation of this phenomenon is not yet available. Fourth, the one-year out-of-sample period constrained the statistical power of the bootstrap tests; while the EMMV significantly outperformed the standard MV, the comparisons with the traditional EWMA-MV and the naïve portfolio did not reach significance at the 5 % level. Longer evaluation windows would provide more definitive statistical evidence. Despite these limitations, the evidence supports the EMMV framework as a practical, data-driven tool for dynamic portfolio selection in nonstationary markets.

4. Conclusion

This paper proposed an EMMV model that constructs portfolio moments by aggregating efficient frontiers from multiple rolling windows with exponential weights. A genetic algorithm with walk-forward cross-validation was employed to select the window size, lag, and forgetting factor directly from the data.

When tested on 30 U.S. stocks from 2010 to 2025, the algorithm identified a window of $m = 83$ days and a low forgetting factor of $\lambda = 0.2344$. In the out-of-sample period (September 2024–September 2025), the EMMV portfolio delivered a buy-and-hold Sharpe ratio of 1.49, more than double that of the equally weighted portfolio and four times that of the static Markowitz model. With quarterly rebalancing and 10 bp transaction costs, the Sharpe ratio remained at 0.69, still exceeding all other optimized benchmarks. The traditional EWMA-MV approach, which smooths individual asset returns, produced concentrated portfolios and lower risk-adjusted returns.

Importantly, the model's performance is robust: a wide plateau of high Sharpe ratios was observed for window sizes between 70 and 90 days and forgetting factors between 0.10 and 0.30. Statistical tests confirmed a significant improvement over the static Markowitz model and a near-significant edge over the equally weighted portfolio. These findings indicate that the EMMV framework, with minimal parameter sensitivity, offers a practical tool for dynamic portfolio selection in non-stationary markets. Future work may extend the approach to multiple asset classes and to adaptive forgetting factors that respond to changing market regimes.

Author Contributions

All authors contributed to the study's conception and design. Data collection and analysis were performed by Kumsong Jo. The first draft of the paper was written by Kumsong Jo and all authors commented on previous versions of the paper. All authors read and approved the final paper.

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Conflict of Interest

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K. Jo, S. Kim & B. Ri

ORCID

Kumsong Jo  <https://orcid.org/0009-0006-6190-3707>

Sunhak Kim  <https://orcid.org/0009-0004-9692-8328>

Bongnam Ri  <https://orcid.org/0009-0000-2648-3264>

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